GIDFINE — PHASE 4 READINESS REVIEW & REPOSITORY AUDIT

Branch: development | **HEAD:** afb4323 | **Stack:** React 19 + TypeScript + Convex | **Spec:** Gidfine_Dev_Guide.md v1.1

1. Repository State & Exact Git Verification

Verification Command	Exact Output / Status
git branch --show-current	development
git rev-parse HEAD	afb43239c112864649e4c63f72d74f8b49daebae
git rev-parse origin/development	afb43239c112864649e4c63f72d74f8b49daebae (Strict match)
git log -10 --oneline	afb4323 (HEAD -> development, origin/development) feat: initialize Convex backend foundation ec023d2 Rename Gidfine_Dev_Guide.md 8d49975 edits reqs b4a488f added the dev guide for the project 7059684 docs: archive VUE_MIGRATION_PLAN.md 62a74ac Complete Vue to React migration f7ed8d1 (origin/main, main) docs: redesign README
git diff main...development	81 files changed, 1897 insertions(+), 10619 deletions(-)
Working Tree Status	Tracked modifications: 7 files (schema, auth, loans, UI wrappers). Untracked: Phase 1-3 migrations, components, services, test suites, and reports.
Main Branch Integrity	**Untouched** at commit f7ed8d1. Zero commits, zero pushes to main.

2. Current Financial Implementation Inventory & Gaps

Financial Dimension	Current Implementation in Repository	Phase 4 Requirement
Financial Ledger	**None exists.** No ledger tables exist in Convex.	Implement append-only ledgerEntries journal with compensating reversals.
Monetary Precision	Integer pesewas (1 GHS = 100 pesewas). Never floats.	Retain strict integer minor units across all financial records.
Rates & Margins	Integer basis points (1% = 100 bps).	Preserve basis points to prevent rounding errors.
Terms Protection	Proposal revision versions & SHA-256 fingerprint.	Re-verify hash at disbursement request and approval to prevent stale terms.
Disbursement State	Read-only check (checkDisbursementReadiness).	Introduce dual-authorization disbursement request, approval, and settlement.
Repayment Allocation	**Zero implementation.** No repayment tables exist.	Implement payment receipt ledger; leave allocation waterfall configurable.

3. Dual-Authorization Disbursement Workflow Architecture

• **Step 1 (Maker Initiation):** Cashier/officer runs checkDisbursementReadiness, confirms verified funding channel, and requests disbursement.
• **Step 2 (Checker Approval):** Agency Admin or Underwriter approves. **Anti-self-approval invariant:** checkerId != makerId is strictly enforced on the server. If the maker attempts approval, the mutation throws FORBIDDEN.
• **Step 3 (Settlement Posting):** Cashier enters verified external transfer reference (bank slip or MoMo ID). Posts to ledgerEntries and advances status to ACTIVE.
• **Real Payments Distinction:** Confirms and records verified external transfer; does not initiate direct bank/MoMo API calls in MVP.

4. Unresolved Business Decisions (Gidfine PRD Section 15)

1. Repayment Waterfall: Priority between arrears penalties, accrued interest, agency margin, and principal reduction is undecided.
2. Interest Calculation Model: Flat rate vs reducing balance vs effective annual rate is explicitly non-binding in PRD Section 6.
3. Prepayment & Penalties: Early settlement interest rebates and grace period policies require product decisions.
Architecture Recommendation: Do not hardcode an arbitrary waterfall. Build an unallocated payment receipt ledger and provide a modular strategy interface once policies are finalized.

5. Phased Implementation Roadmap for Phase 4

Sub-Phase	Focus Area	Deliverables
Phase 4A	Ledger Schema & Idempotency	ledgerEntries, disbursementRequests, idempotency index.
Phase 4B	Dual-Auth Disbursement	requestDisbursement, approveDisbursement (maker-checker).
Phase 4C	Compensating Reversals	reverseLedgerEntry, getApplicationLedger queries.
Phase 4D	Repayment Receipts	recordRepayment, receipt tracking, modular allocation.
Phase 4E	UI Integration	CashierDisbursementManager.tsx, LoanLedgerView.tsx.
Phase 4F	Testing & Verification	Concurrency tests, maker-checker tests, build verification.

6. Summary & Current Status

Current Status: No files have been committed, pushed, or deployed. All 30 Node unit tests, 6 Python regression tests, and full production build remain passing with 0 errors. Awaiting explicit user approval before beginning Phase 4 implementation.